

Space Age

When it comes to the universe, infinity is just the beginning. Look at some of the largest things it has to offer.

Meet Angelina: An AI that thinks

A computer scientist's PhD involves developing an AI that can think creatively on its own and design games: <http://dgit.in/angelAI>

THE RISE OF BIONICS

"By the late twentieth century, our time, a mythic time, we are all chimeras, theorized and fabricated hybrids of machine and organism; in short, we are cyborgs."

Donna J. Haraway, Author

Sanket Chaukiyal

readersletters@thinkdigidit.com

Although a few years late, Donna J. Haraway's statement couldn't have been more prophetic. We have moved more closer to an fusion of man and machine than one could have imagined all those years ago. Suddenly, Robocop and Bionic Woman don't seem so distant anymore.

Let us take a look at the world of artificial body parts and exo-skeletons that exists out there.

PROSTHETICS

Prosthetics have come a long way from the pirate days of wooden leg stumps to thought-controlled, touch-sensitive, 3D-printed limbs of today. They are lighter, built with better material and come close to looking and feeling like the real thing. Some prosthetics enable a natural walk while some can even turn a person into a speedster. The real difference comes when the prosthesis gives the user exactly what they want. There are some innovations in prosthetics that come close to the real thing (till we figure out how to regenerate like reptiles without turning into a giant human-lizard hybrid with a superiority complex). People are developing a host of new technology for prosthetics like mind-controlled prosthetics, touch sensitive prosthetics and – believe it or not – app controlled prosthetics. Way to go, smartphone generation.

3D-printed prosthetics

Although light years behind the Star Trek replicator, the 3D printer is definitely a step closer. Very much in rage now, the 3D printer allows us to create (read: print) very much everything that

we have the design for (remember the 3D printed larynx of a velociraptor in Jurassic Park 3?), the printing material isn't a limit either, as we will soon see.

Prosthetic hand

At a young age of 17, Easton LaChapelle developed a 3D-printed prosthetic arm that can be made for under \$500 (a state-of-the-art prosthetic can cost around \$80,000). This arm is so good that it got him a meeting with Barack Obama and then got him into NASA's Robonaut Project (a humanoid robot created to assist astronauts in space). The arm weighs around 10 pounds, uses an EEG headband for controlling its movements and is easy to put on and take off, unlike other prosthetics which are very heavy or require invasive neural surgery for user interface.

Prosthetic jaw

Prosthetics are not just limited to hands and legs, it all depends on creativity and technology. An 83-year old lady got implanted with a titanium (that's right) lower jaw as her infected jaw was beyond healing. The prosthetic jaw implant was so successful that the lady could move the jaw within a day of the surgery, though the healing process took longer. Researchers and engineers from Belgium and the Netherlands printed the jaw using titanium powder and sprayed it with artificial bone. It's sad about her old jaw, but on the bright side: walnuts won't be a problem for the lady anymore. Neither will self-defence, actually.

Got music in your bones?

Being humans by nature, our thoughts often span out beyond the normal and this in turns churns out some unusual results. Although not benefiting in anyway by means of physical body enhancements or support, the musical prosthetics created by two Music PHD students from McGill University, Montreal are extraordinary. These 3D-printed and laser-cut prosthetics include spines, ribs, etc and play music when the person wearing them dances. They are embedded with sensors, power supply, wireless transmitters, etc. that allow the wearer to play music by touch and

